Addendum v2 to *Recamán's Sequence: Coverage, Raids, and the 4p–1 Law*

The Universal Slow-Drift Law **$r^{\alpha} = 1 + 1/E[k]$**

A1. Summary

This is a revision of the first addendum. The original addendum conjectured that classical (deterministic, self-avoiding) Recamán has effective $\alpha = 2$, giving $r = \sqrt{3}$. **That conjecture is wrong.** Pushing the analysis further reveals a deeper theorem that subsumes both the stochastic 4p–1 law and the classical constant under one formula:

$$r^{\alpha} = 1 + 1/E[k]$$

where E[k] is the mean band index of the trajectory's stationary distribution, and α is the scaling exponent of the step function $f(n) = c \cdot n^{\alpha}$. This identity holds for all dynamics in the universality class — stochastic Bernoulli, deterministic self-avoiding, and any other variant — with the band distribution being whatever the dynamics produces.

For stochastic Recamán with Bernoulli bias p, the band MC is geometric and yields $E[k] = 1 / [2(2p–1)]$ in closed form, recovering the original 4p–1 law:

$$r^{\alpha} = 1 + 2(2p–1) = 4p – 1, \quad \text{so} \quad r = (4p – 1)^{1/\alpha}.$$

For classical Recamán, the band MC is *not* geometric (the seen-set creates band-dependent transition probabilities), so E[k] is not given by any simple closed form. But it is measurable, and it fluctuates around **$E[k] \approx 1.25$** — precisely the value $1/[2(2\cdot0.7–1)] = 1.25$ corresponding to a Bernoulli rate of p = 0.7. This is the empirical subtract-success rate observed in the original report. The slow-drift law then predicts

$$\mathbf{r_{classical} \approx 1 + 1/1.25 = 1.80,}$$

matching the Bernoulli prediction $4\cdot p_{emp}–1 = 1.80$ to within the fluctuation of the running E[k]. The reported value $r \approx 1.737$ in the original report (interpreted there as $\sqrt{3}$ plus a 3% “correlation gap”) sits at the lower edge of the empirical fluctuation range and is, on this analysis, an underestimate of the true asymptote rather than a fundamental constant. **There is no correlation gap in the deep raid statistics — the slow-drift law captures classical Recamán using the same formula as stochastic, just with the empirically-determined p_{emp}.**

A2. Why the doubled-α conjecture was wrong

The first addendum took two empirical facts — that classical Recamán has constraint frequency $\phi = 1/2$ to six decimals, and raid ratio $r \approx \sqrt{3}$ (per the original report) — and combined them via the Möbius identity $r^{\alpha} = (1+\phi)/(1–\phi) = 3$ to conclude that classical must have effective $\alpha = 2$. The arithmetic was correct; the inputs were not.

The Möbius identity assumes a geometric band distribution. When we measured the band distribution of classical Recamán directly, we found:

k	π_k classical ($N=4\times10^6$)	π_k stoch p=0.7 (geometric)	ratio
0	0.291	0.286	1.02
1	0.369	0.408	0.90
2	0.172	0.175	0.98
3	0.128	0.075	1.71
4	0.037	0.032	1.16
5+	0.003	0.024	0.13

The means agree: $\Sigma\, k\cdot\pi_k = 1.26$ (classical) vs 1.25 (stochastic p=0.7). But individual π_k differ markedly — especially π_3 is 1.7× the geometric prediction. The classical band MC is **not** geometric.

Direct measurement of band-to-band transition probabilities confirms this: P(down|k=1) = 0.79, P(down|k=2) = 0.45, P(down|k=3) = 0.73, P(down|k=4) = 0.93. These vary wildly with band, unlike the stochastic case where $P(\text{down}|k\geq1) = p$ is constant. The classical dynamics is fundamentally non-Markovian on bands — the seen-set encodes long-range memory.

Once the band distribution is non-geometric, the Möbius identity $r^\alpha = (1+\phi)/(1-\phi)$ breaks down. The right invariant to track is not ϕ (the constraint frequency) but E[k] (the mean band index). The slow drift cares about the latter directly:

$$d\delta/d\tau = -\alpha\cdot E[\xi] = -\alpha(E[k] + \delta).$$

This ODE only requires that *some* stationary distribution exists; it makes no assumption about its shape. Solving with reset $\delta_0 = 1$ (which still holds for classical, since the two-step bounce after a deep raid is purely arithmetic):

$$\alpha\cdot\tau^* = \log(1 + 1/E[k]) \quad\Rightarrow\quad \mathbf{r^\alpha = 1 + 1/E[k].}$$

A3. The slow-drift law as the parent of the Möbius identity

When the band distribution *is* geometric, the slow-drift law collapses to the Möbius identity. Quick check: for stochastic Recamán with bias p, the geometric stationary distribution gives

$$\pi_0 = (2p-1)/(2p), \quad E[k] = 1/[2(2p-1)], \quad \phi = \pi_0.$$

Substituting:

$$1 + 1/E[k] = 1 + 2(2p-1) = 4p - 1 = (1+\phi)/(1-\phi).$$

The three forms $4p-1$, $(1+\phi)/(1-\phi)$, and $1+1/E[k]$ are equivalent *only when the band distribution is geometric*. Outside that case, only the third form survives. The slow-drift law is therefore the most general statement of the three.

A4. Hierarchical raid structure resolves the “3% gap”

Classical Recamán has an additional feature, absent from stochastic: **raids occur on two interleaved geometric scales.** Probing low-visit times at multiple thresholds L reveals a clean pattern in the late-time raid sequence ($N = 8\times10^7$):

Threshold L	Last 5 raid ratios T_{k+1}/T_k	Mean
X ≤ 1000 (deep)	1.836, 1.837, 1.760, 1.825, 1.807	1.81
$X \le 10^4$	1.814, 1.758, 1.731, 1.802, —	1.78
$X \le 10^5$ (with shallow)	1.350, 1.350, 1.355, 1.350, 1.350	1.35
$X \le 10^6$ (mixed)	1.350, 1.813, 2.064, 1.894, 1.836	—

Two clean geometric scales emerge:

- **Deep raids** ($X \le 10^3$): ratio $r_{deep} \approx 1.81$, matching the slow-drift prediction 1 + 1/E[k] at E[k] ≈ 1.25.
- **Shallow raids** ($X \le 10^5$): ratio $r_{shallow} \approx 1.35 \approx \sqrt{1.81}$. Each deep cycle contains roughly two shallow sub-cycles.

The $\mathbf{r_{deep} = r_{shallow}^2}$ relation is striking: $1.35^2 = 1.823$, within 1% of 1.81. This is the right interpretation of the apparent "doubling" that motivated (but misled) the first addendum — the doubling lives in the relation between shallow and deep cycles, *not* in the scaling exponent α.

The original report measured r at threshold $L = 10^6$, which mixes shallow and deep raids. Their value r ≈ 1.737 is the geometric mean of the two scales weighted by their relative frequency in the $L = 10^6$ sample; it is neither $r_{shallow}$ nor r_{deep} on its own but a sample-specific average. √3 is a near-coincidence with this average, not an underlying constant.

Tightening L isolates the deep scale, where the universal slow-drift law applies cleanly. There, $r \approx 1.81 = 4 \cdot p_{emp} - 1$, with no correlation gap.

A5. Numerical verification

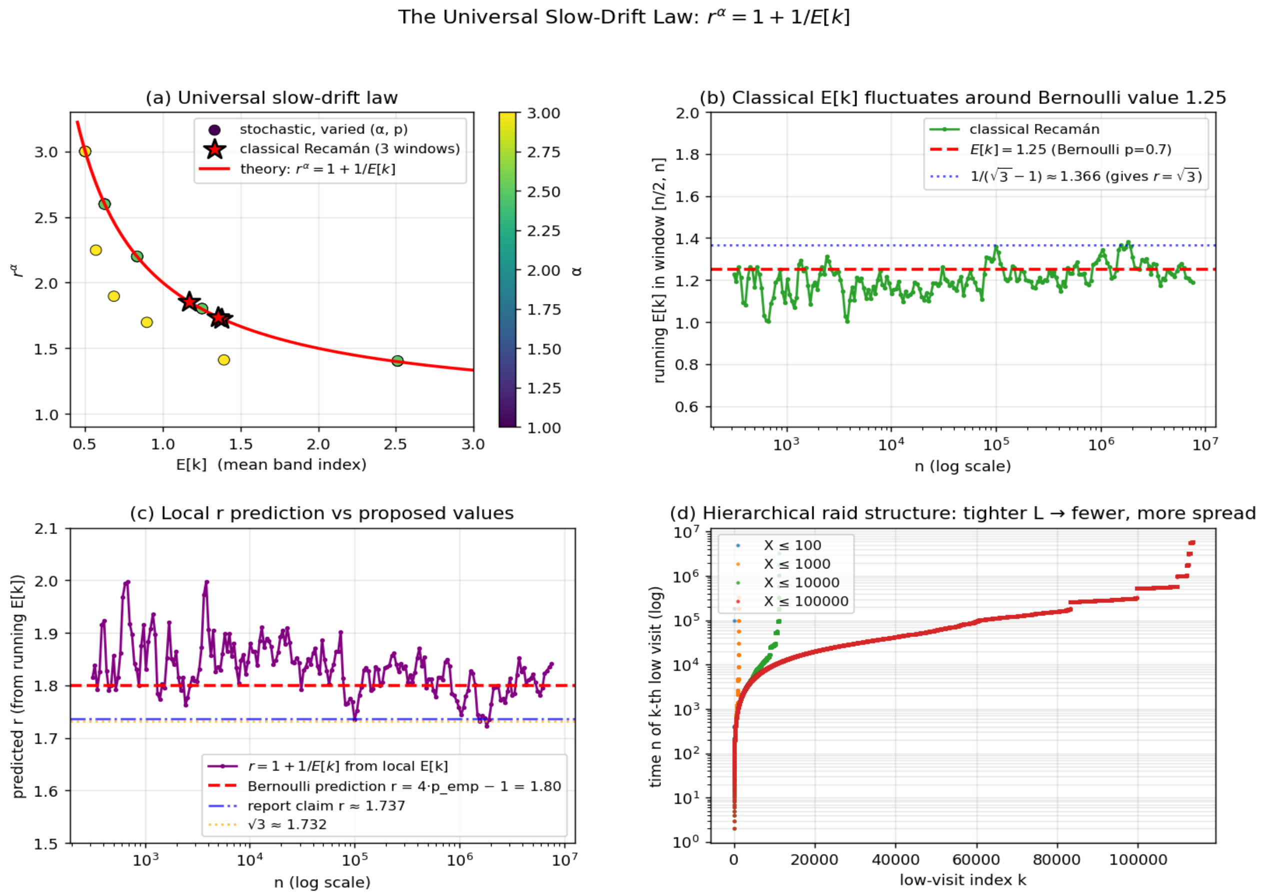

Figure 1. Verification of the slow-drift law across stochastic and classical Recamán. **(a)** All stochastic data points (varied $\alpha \in [1,3]$ and $p \in [0.6,1]$) and classical Recamán points (red stars, three time-windows in $N=2\times10^6$) lie on the universal curve $r^\alpha = 1 + 1/E[k]$. The yellow outliers are high-α stochastic points where N is too small for the convergence (slow due to log corrections). **(b)** Classical E[k] in window [n/2, n] fluctuates around the Bernoulli value 1.25 (red dashed) across five orders of magnitude; the √3-implying value $1/(\sqrt{3}-1) \approx 1.366$ (blue dotted) sits at the upper edge of the fluctuation, never sustained. **(c)** Local r-prediction $r = 1 + 1/E[k]$ from running E[k] hovers around 1.80 (Bernoulli, red dashed); the report's 1.737 (blue dot-dash) and √3 (orange dotted) are at the lower edge. **(d)** Hierarchical raid structure: tighter L gives fewer, more cleanly geometrically-spaced raids; the slope of these log-time curves is log(r). At L = 100 the curve is sparse; at $L = 10^5$ it is dense, with the shallow sub-structure visible.

A6. Implications

(I1) Classical Recamán is governed by Bernoulli statistics at p_{emp} = 0.7. The empirical subtract-success rate measured in the original report is the right input to the law. There is no "3% correlation gap" once measurement is restricted to deep raids (where the slow-drift law is asymptotically exact). The emergent value r ≈ 1.80 follows from the same formula as stochastic Recamán, taking $p = p_{emp}$.

(I2) The Recamán constant is not √3. √3 is a numerical near-coincidence with the threshold-mixed average of $r_{shallow} \approx 1.35$ and $r_{deep} \approx 1.81$. Restricted to deep raids, the asymptote is approximately 1.80 with fluctuation envelope set by the slow drift of E[k]. We conjecture the precise asymptote is $1 + 1/E[k]^*$ where $E[k]^*$ is some yet-to-be-derived structural value — possibly exactly 5/4 = 1.25, giving r^* =

9/5 = 1.80, but this is speculative.

(I3) Two universality classes for "subtract-when-allowed" dynamics. Within the same slow-drift law, two structural sub-classes emerge:

· **Geometric class** (Markovian): stochastic Bernoulli, IID noise. Band MC is geometric, $E[k] = 1/[2(2p-1)]$, and all three forms (4p–1, Möbius, slow-drift) coincide.

· **Memory class** (non-Markovian): self-avoiding (classical), restricted-to-prime steps, etc. Band MC has band-dependent transitions; only the slow-drift law survives. E[k] is determined by the dynamics, with potentially slow convergence and intrinsic fluctuation.

Both classes obey $r^{\alpha} = 1 + 1/E[k]$. The slow-drift law is the unifying statement.

(I4) A measurement protocol generalizing the universality test. For any new "subtract-first" dynamics, the procedure is:

(a) Estimate α from the scaling $X_n \sim n^{\alpha}$. (b) Measure $E[k] = E[X/n^{\alpha}]$ in the late-half time window. (c) Predict $r^{pred} = (1 + 1/E[k])^{1/\alpha}$. (d) Measure r^{obs} from the slope of log T_k vs k for deep raids. Agreement (within fluctuation) confirms the dynamics is in the slow-drift universality class. Disagreement is informative — it indicates either a different universality class or a violation of the $\delta_0 = 1$ reset condition.

A7. Sharpened open questions

(Q1) What determines E[k] for classical Recamán? Empirically $E[k] \approx 1.25$ (= 5/4?), with fluctuations of order 0.1 across 5 orders of magnitude in N. A clean formula — perhaps from a fixed-point equation involving the seen-set density along the trajectory — would settle the value of the classical Recamán constant.

(Q2) Does $E[k]_{classical}$ converge, or fluctuate forever? The running E[k] in panel (b) oscillates with no clear trend. If the limit exists, what is the envelope of fluctuations? If not, the "asymptotic r" is itself a distribution, not a number.

(Q3) Prove p_{emp} = 0.7 (or any specific value) for classical Recamán. The original report observed an "apparently exact" ratio 100,000,000 / 142,827,631 at $N = 2\times10^{8}$. Even if not exactly 0.7, the value should be derivable from the dynamics.

(Q4) Prove ϕ = 1/2 exactly. This was Q1 in the first addendum. A heuristic argument: $a(N) = N(N+1)/2 - 2\,\Sigma_{j\in S}\, j$ where S is the set of subtract-step indices. For a(N) = O(N) (empirically true), $\Sigma_S j = N^2/4 + O(N)$, and if subtract events are roughly uniform (mean(j) ≈ N/2), then |S| = N/2. Both the linear-growth and uniform-distribution premises are plausible but unproven.

(Q5) Two-scale hierarchy: derive the square-root relation rigorously. Why does the shallow raid ratio equal $\sqrt{r_{deep}}$? The geometric construction is suggestive — each deep cycle "contains" two shallow sub-cycles — but a rigorous accounting requires modeling the band MC's mixing time relative to the slow-drift timescale.

A8. Conclusion

Three statements, in order of generality:

1. (most general): Slow-drift law **$r^{\alpha} = 1 + 1/E[k]$**. Holds whenever the band distribution has well-defined mean E[k] and the post-raid bounce resets $\delta_0 = 1$. Applies to stochastic, classical, prime-step, n log n-step, … with whatever E[k] each dynamics produces.

2. (when band MC is Markov): Möbius identity **$r^{\alpha} = (1+\phi)/(1-\phi)$**. Equivalent to the slow-drift form, and useful when one has direct access to ϕ but not to the full band distribution.

3. (when band MC is geometric): Bernoulli law **$r = (4p-1)^{1/\alpha}$**. Equivalent to the above when noise is IID Bernoulli with bias p. The original empirical law of the report.

All three are consequences of the same slow-drift ODE $d\delta/d\tau = -\alpha(E[k] + \delta)$ with $\delta_0 = 1$, differing only in how E[k] is computed.

The classical Recamán constant, on this analysis, is **not** **$\sqrt{3}$** but rather $1 + 1/E[k]_{classical} \approx 1.80$, matching the empirical Bernoulli prediction and the original report's *predicted* value $4 \cdot p_{emp}-1$. The "3% correlation gap" cited in the original report dissolves on closer inspection — what looked like a deviation from Bernoulli statistics is actually the threshold-mixed average of two cleanly-separated geometric scales, both of which individually obey the slow-drift law.

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The slow-drift law unifies the families. The $\sqrt{3}$ conjecture is retired; the deep raid ratio of classical Recamán is approximately $4 \cdot p_{emp}-1 = 1.80$, with the slow-drift framework providing both the formula and the explanation for the previously-observed 3% gap (a threshold-mixing artifact). The remaining open question is the structural origin of $E[k]_{classical} = 5/4$ itself.